

AM fungal mutualism disruption dictates native plant photosynthetic responses to *Alliaria petiolata*

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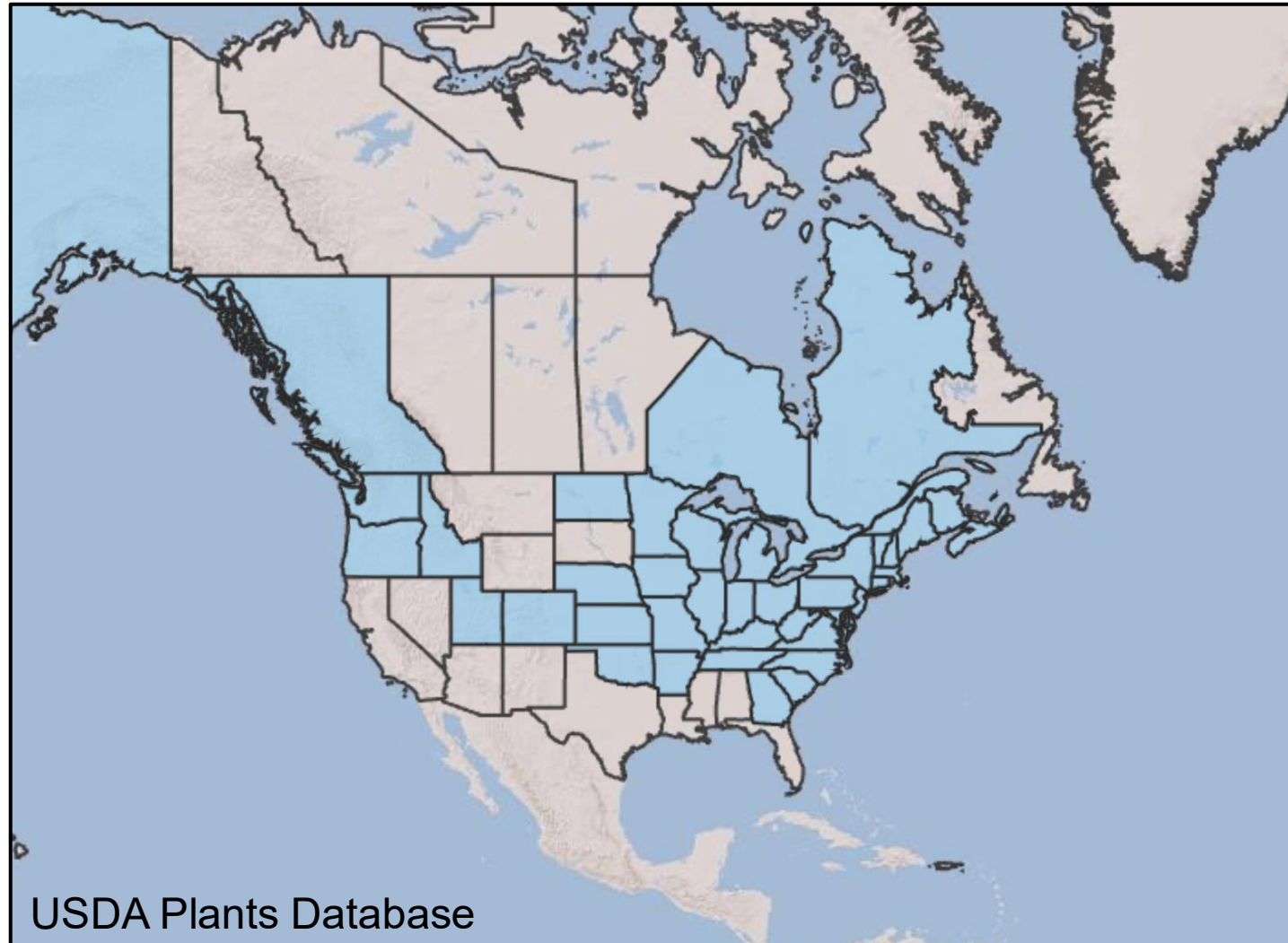
TEXAS TECH
UNIVERSITY.



Swiss Federal Institute for Forest,
Snow and Landscape Research WSL



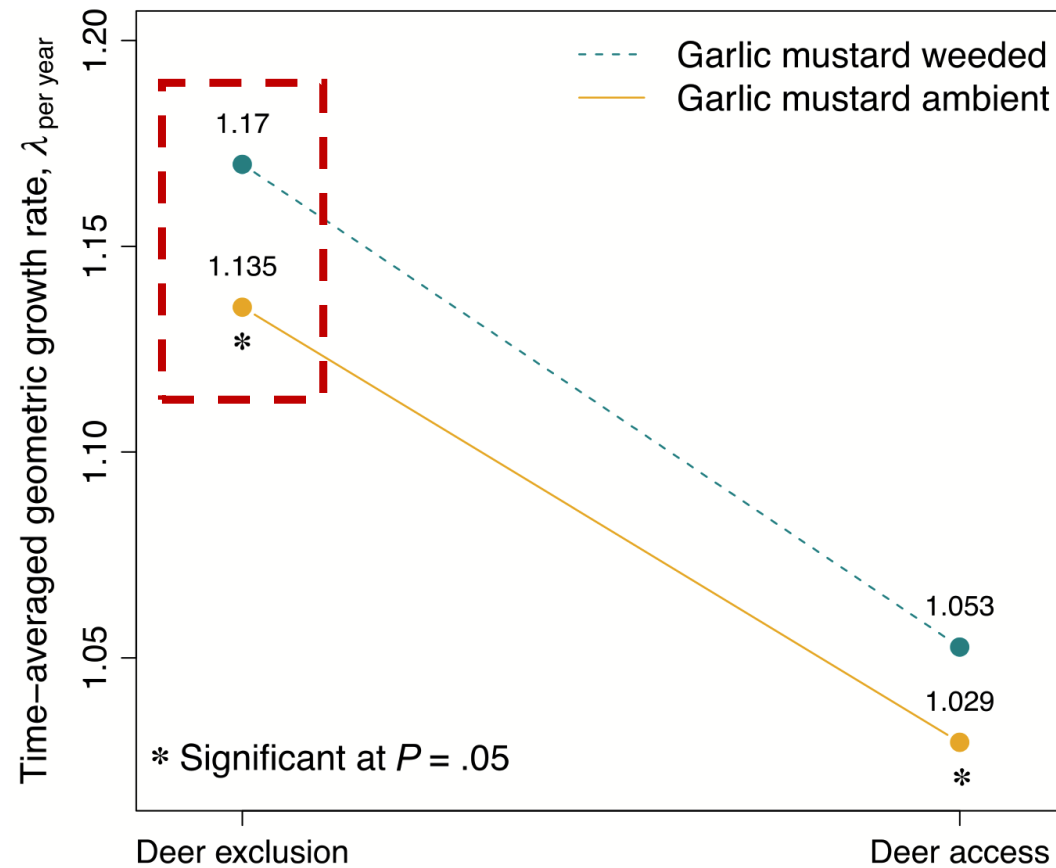
Garlic mustard (*Alliaria petiolata*, Brassicaceae) is a widespread allelopathic invader in North America



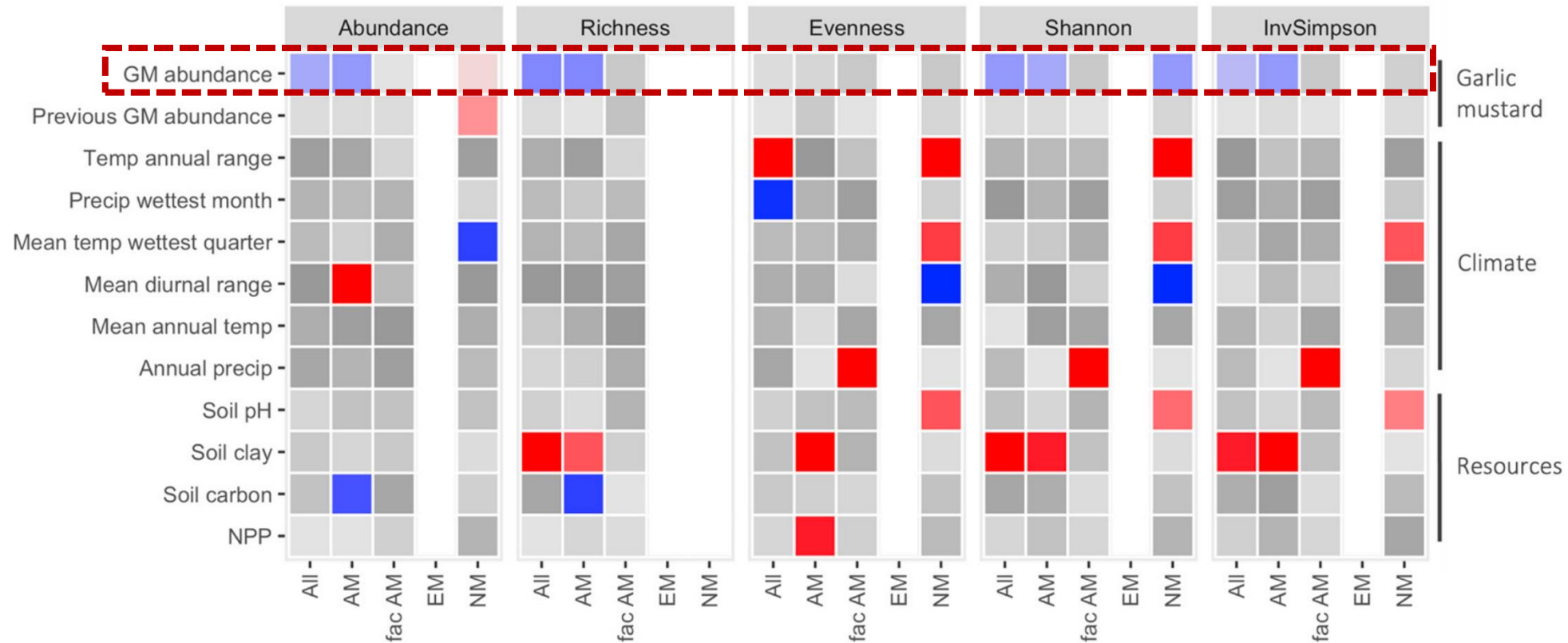
Biennial understory species from Eurasia

Frequently documented ability to release antimicrobial compounds into soils

Garlic mustard negatively affects native plant community growth rates



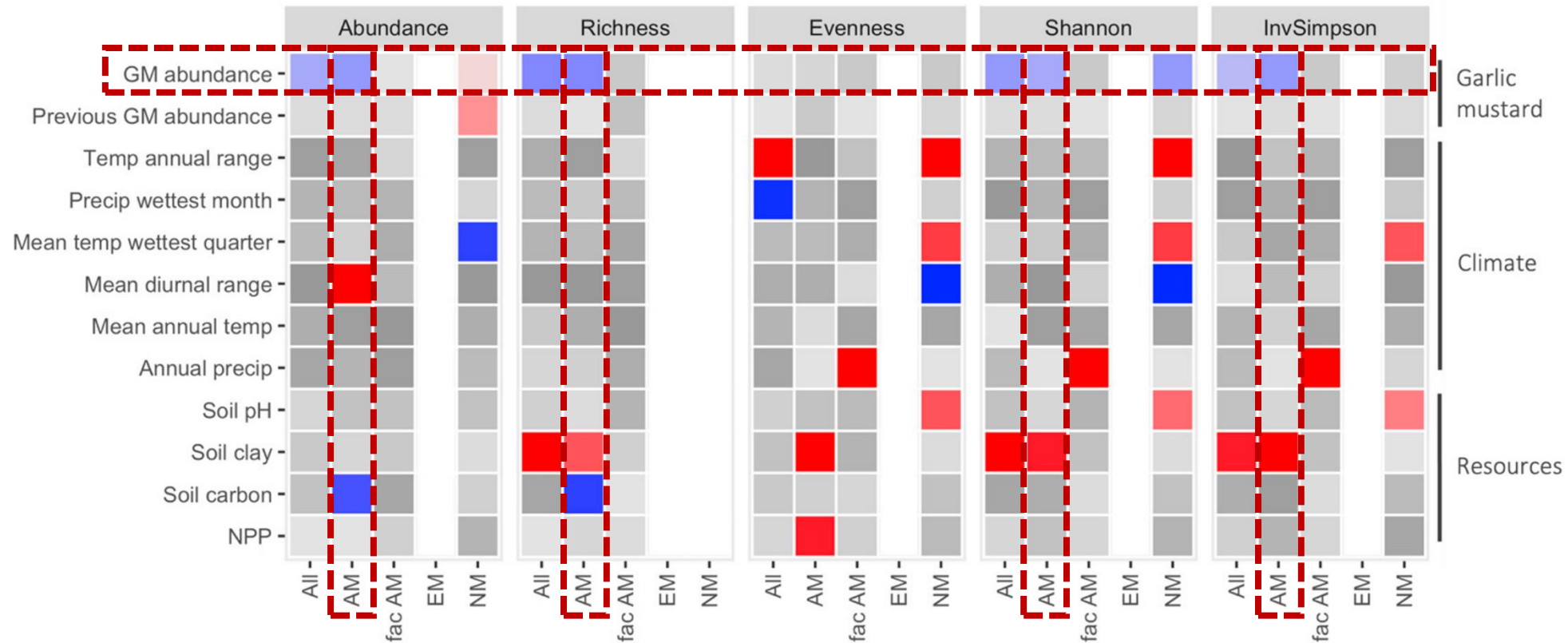
Garlic mustard presence decreases native plant community biodiversity



Blue Negative effect

Red Positive effect

Garlic mustard presence decreases native plant community biodiversity*

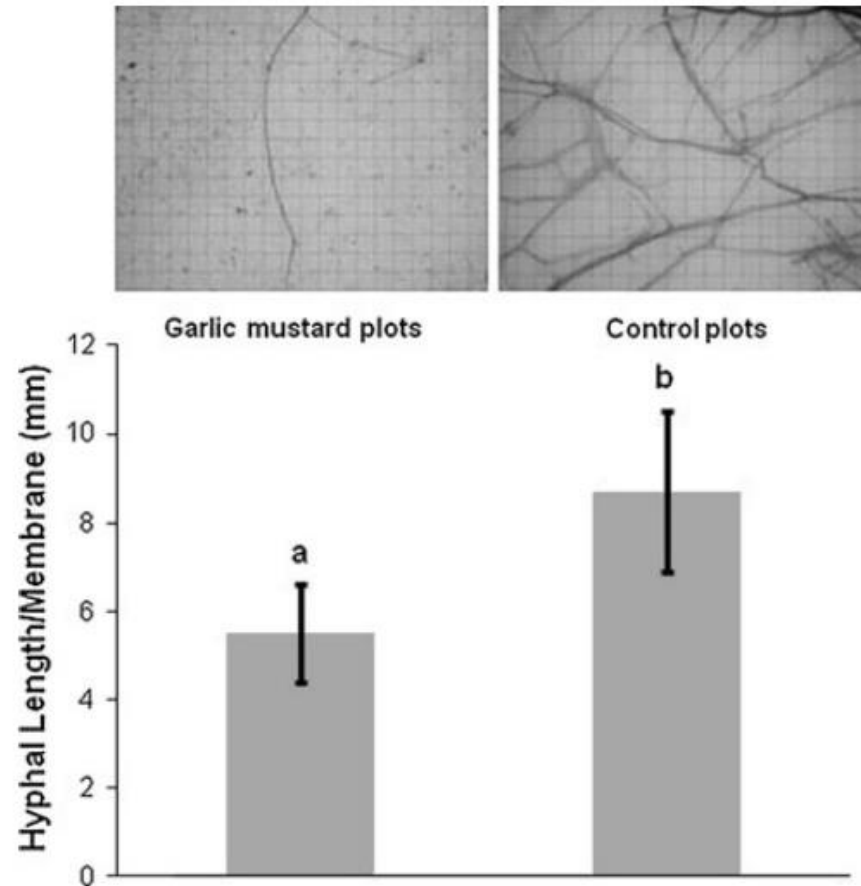


Blue Negative effect

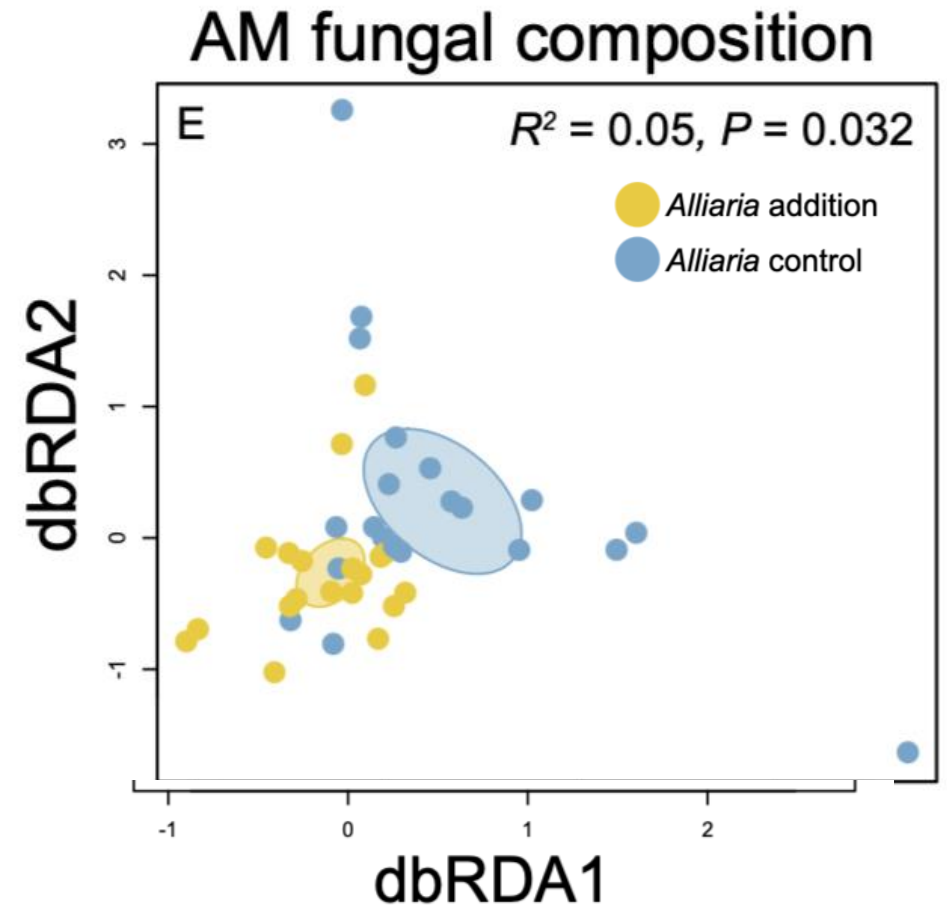
Red Positive effect

*general patterns explained by species that form associations with arbuscular mycorrhizal (AM) fungi

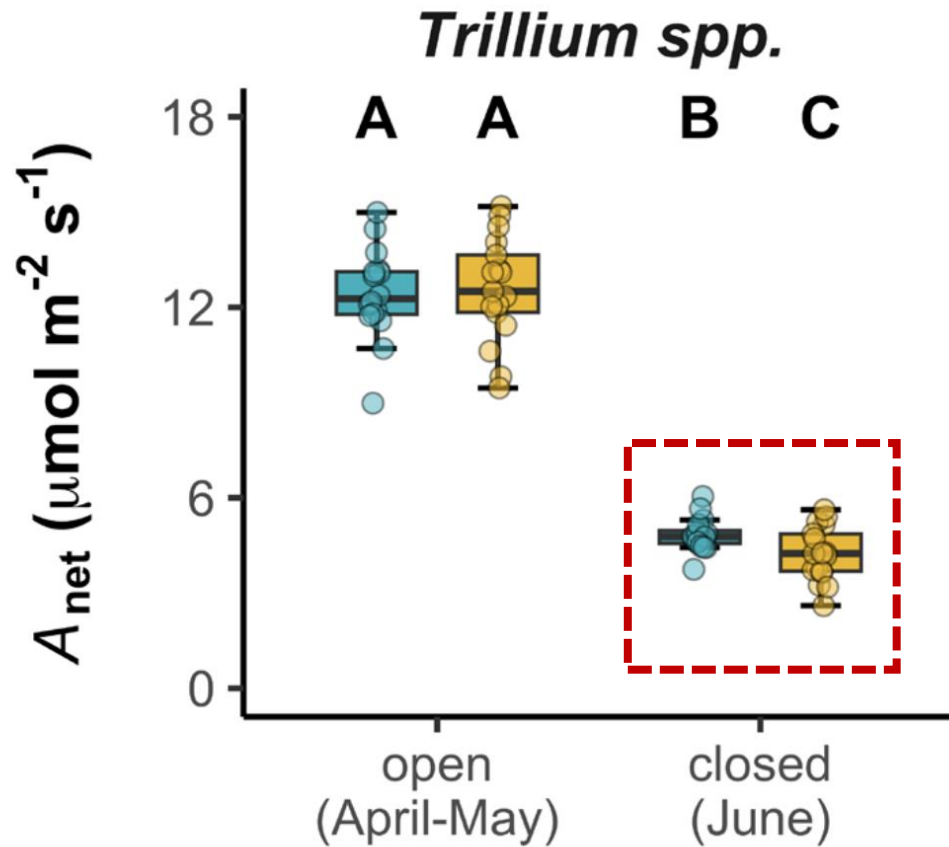
There is ample evidence showing that garlic mustard disrupts AM fungal communities



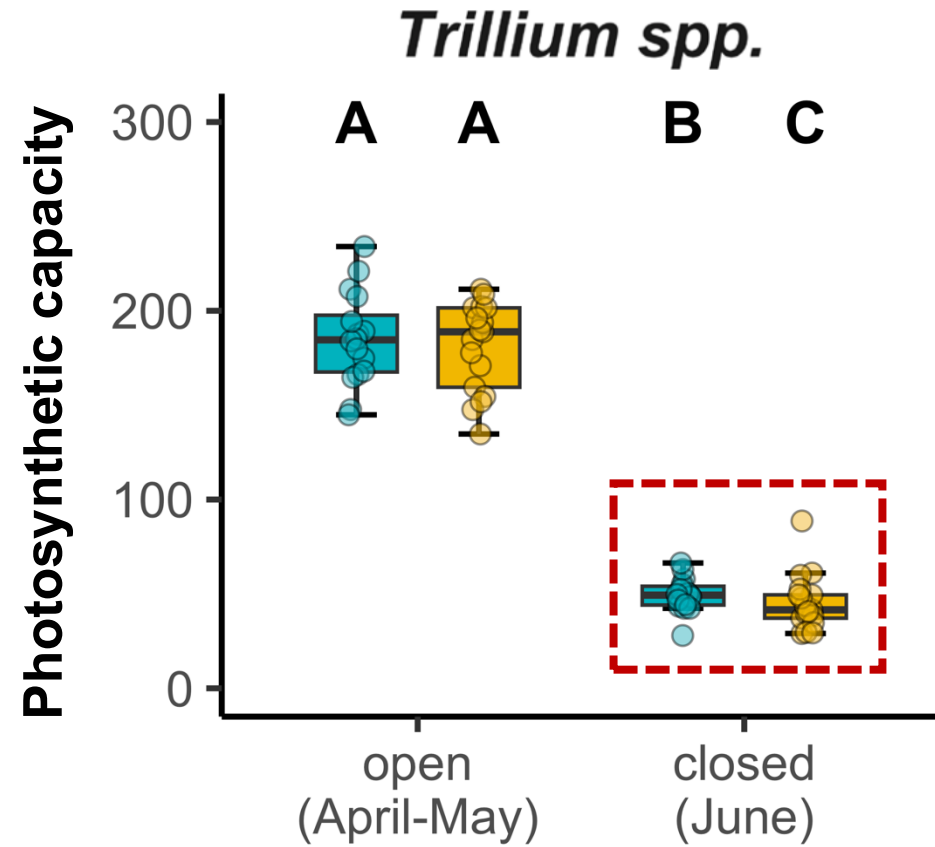
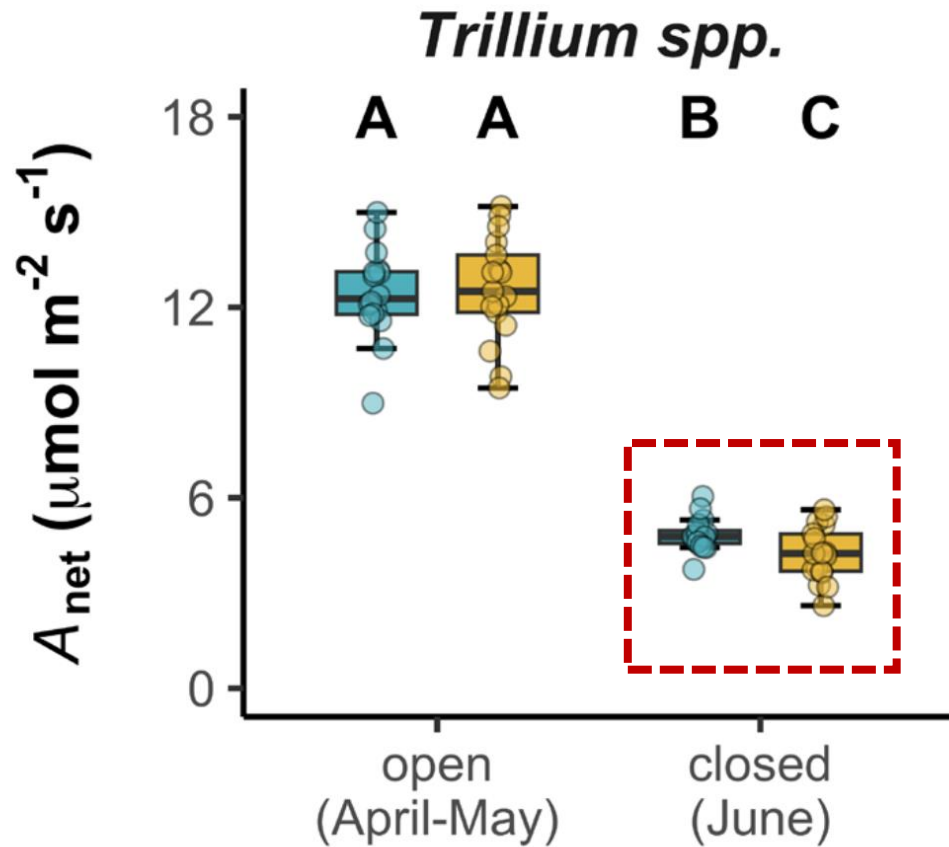
Alliaria treatment



Recent evidence shows that garlic mustard also decreases native plant photosynthesis

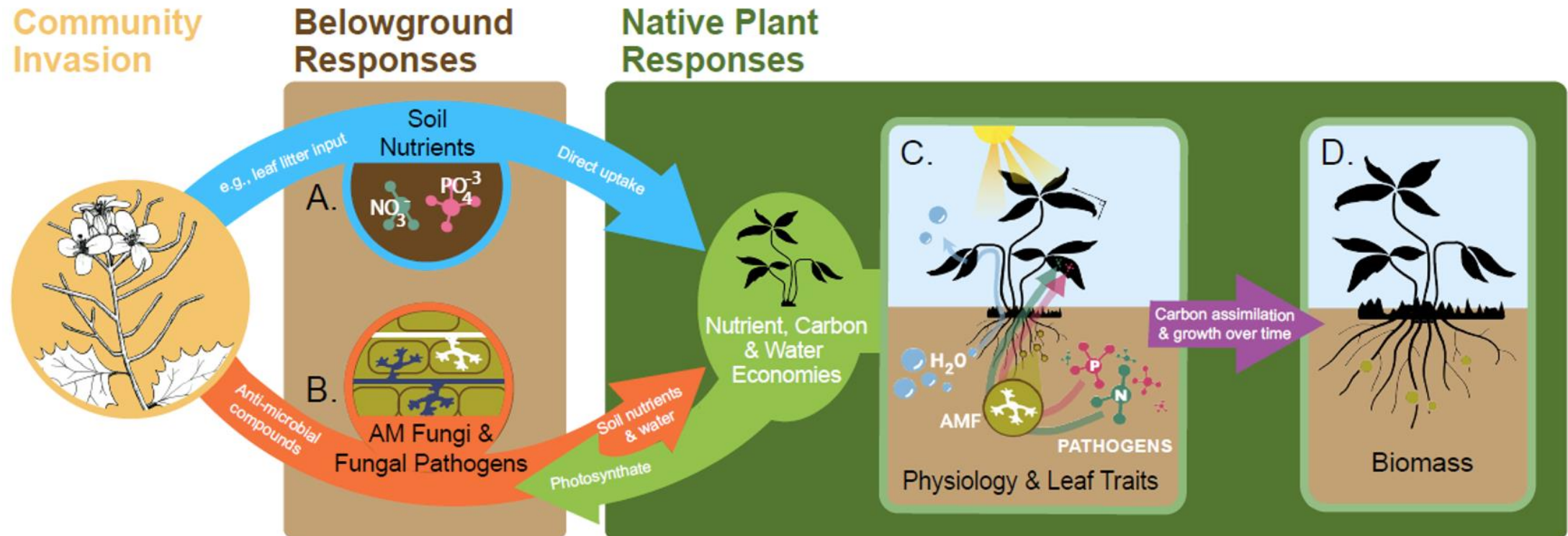


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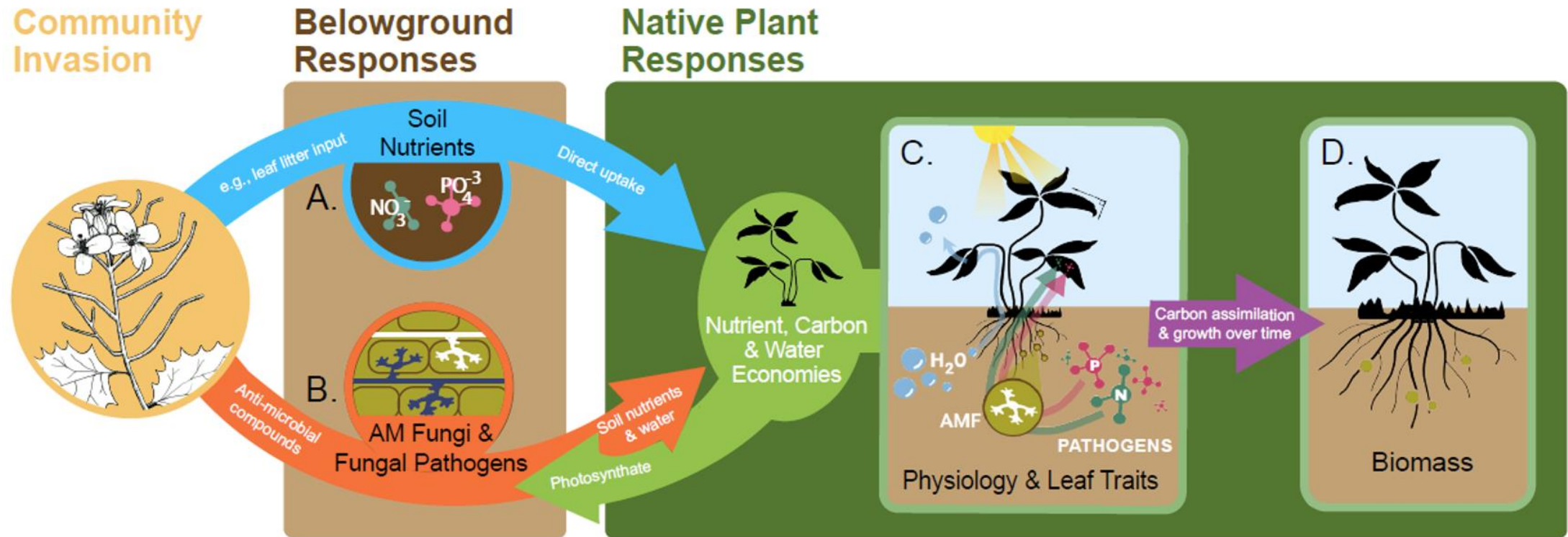


*often through changes in photosynthetic capacity

Photosynthetic responses to garlic mustard are likely driven by changes in AM mutualism breakdown



Photosynthetic responses to garlic mustard are likely driven by changes in AM mutualism breakdown



Photosynthetic responses to garlic mustard have not yet been causally linked to AM fungal mutualism breakdown

Study Questions

Are photosynthetic responses to garlic mustard driven by AM fungal mutualism disruption?

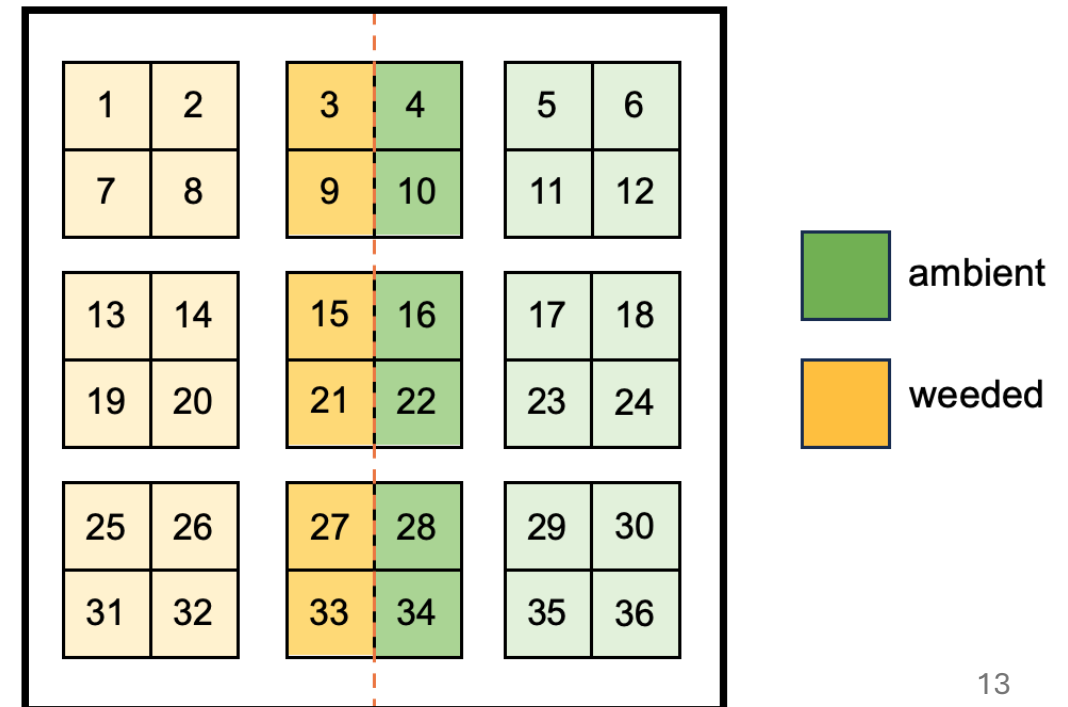
Study Questions

Are photosynthetic responses to garlic mustard driven by AM fungal mutualism disruption?

What is the mechanism that explains AM fungal effects on physiological responses to garlic mustard?

Long-term garlic mustard removal experiment

- Fox Chapel, Pennsylvania
- Fencing established in 2002 to ward out deer/macro-herbivores
- Garlic mustard weeding treatments began in 2006
 - Split plot design



Plant-soil feedback experiment

We harvested *Trillium* spp. and collected soils growing in long-term experiment and imposed the following treatments in split-pots

Plant legacy treatment

Trillium spp. with history of growing in ambient or weeded field treatment

Soil source

Bulk soil collected from ambient or weeded plots

AM fungal source

AM fungal inoculum collected from ambient or weeded plots

Plant-soil feedback experiment took place in growth chambers under conditions that replicated field conditions



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Measurements

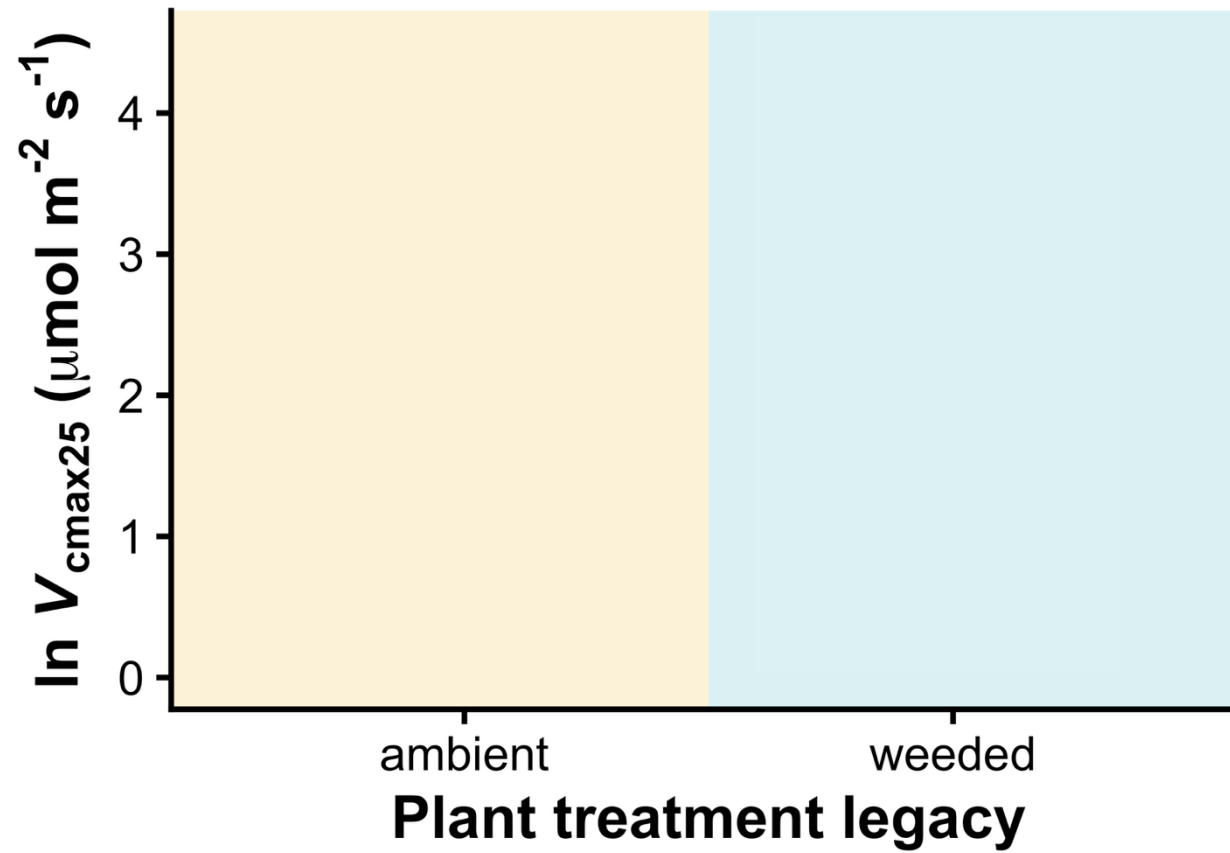
Measurement category	Approach	Rationale/Reason for measurement
Leaf-level photosynthesis	CO ₂ response curves	To understand how treatments modified net photosynthesis, stomatal conductance, and photosynthetic capacity

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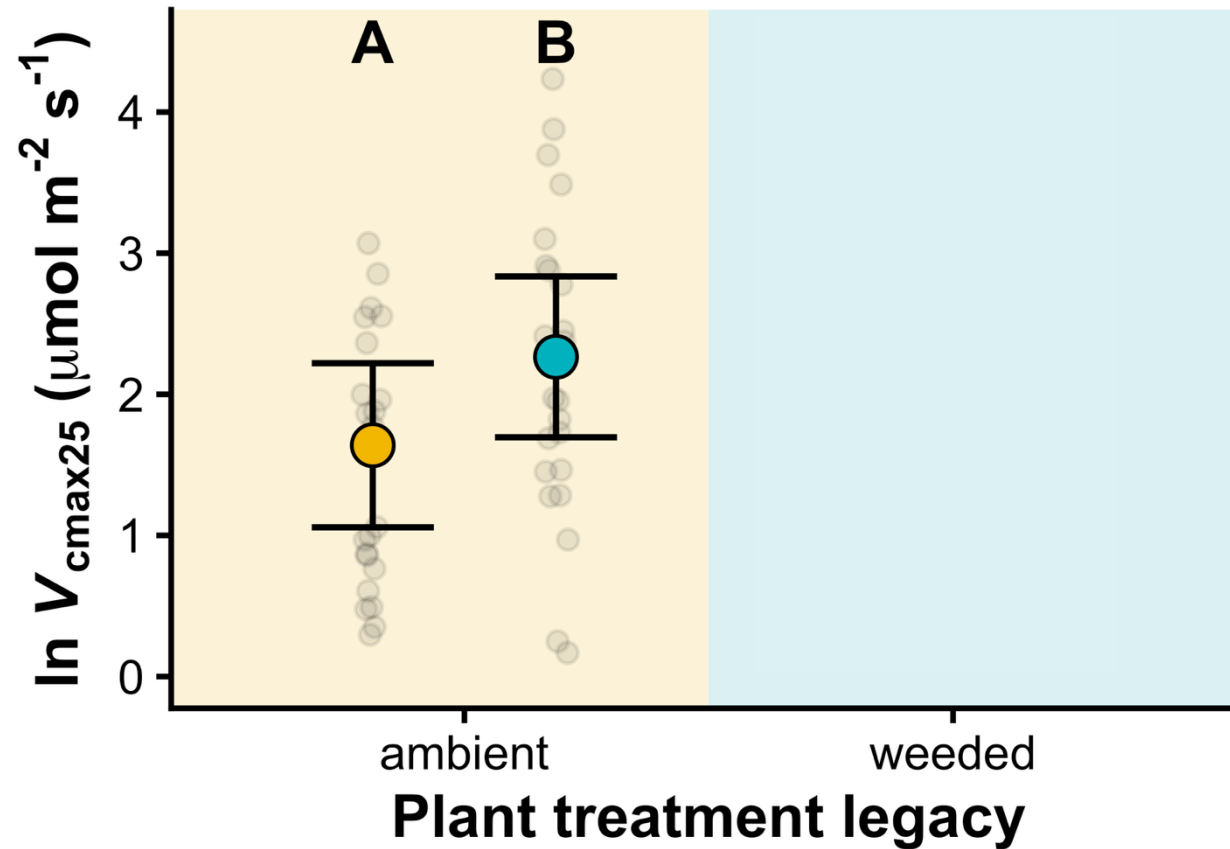
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AM fungal community	Bulk soil	Quantify AM asseptate fungal hyphal lengths in soil to understand how treatments modified AM fungal biomass, connect with physiology
	 Coming soon: also from roots!	



Ambient = GM present
Weeded = GM absent

AM fungal source ● ambient ● weeded

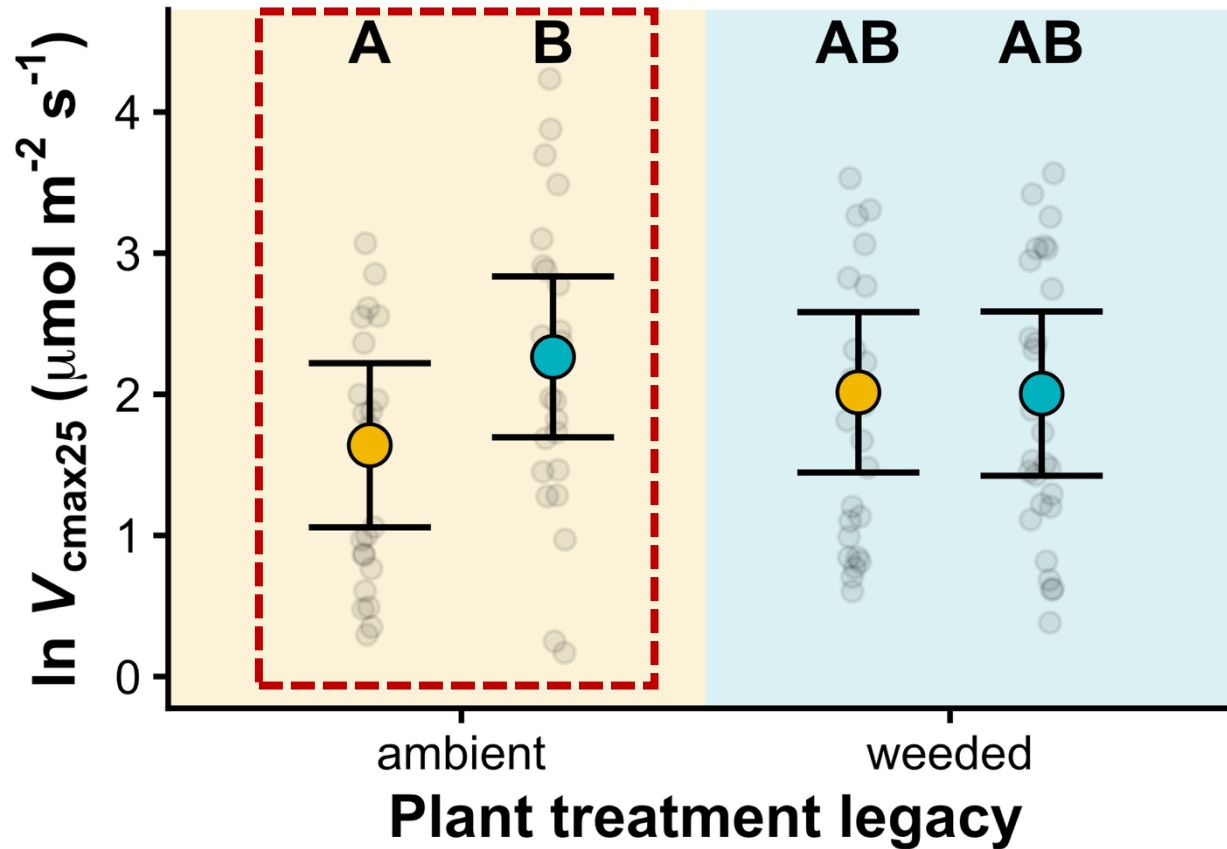
Ambient AM fungal source significantly decreases photosynthetic capacity



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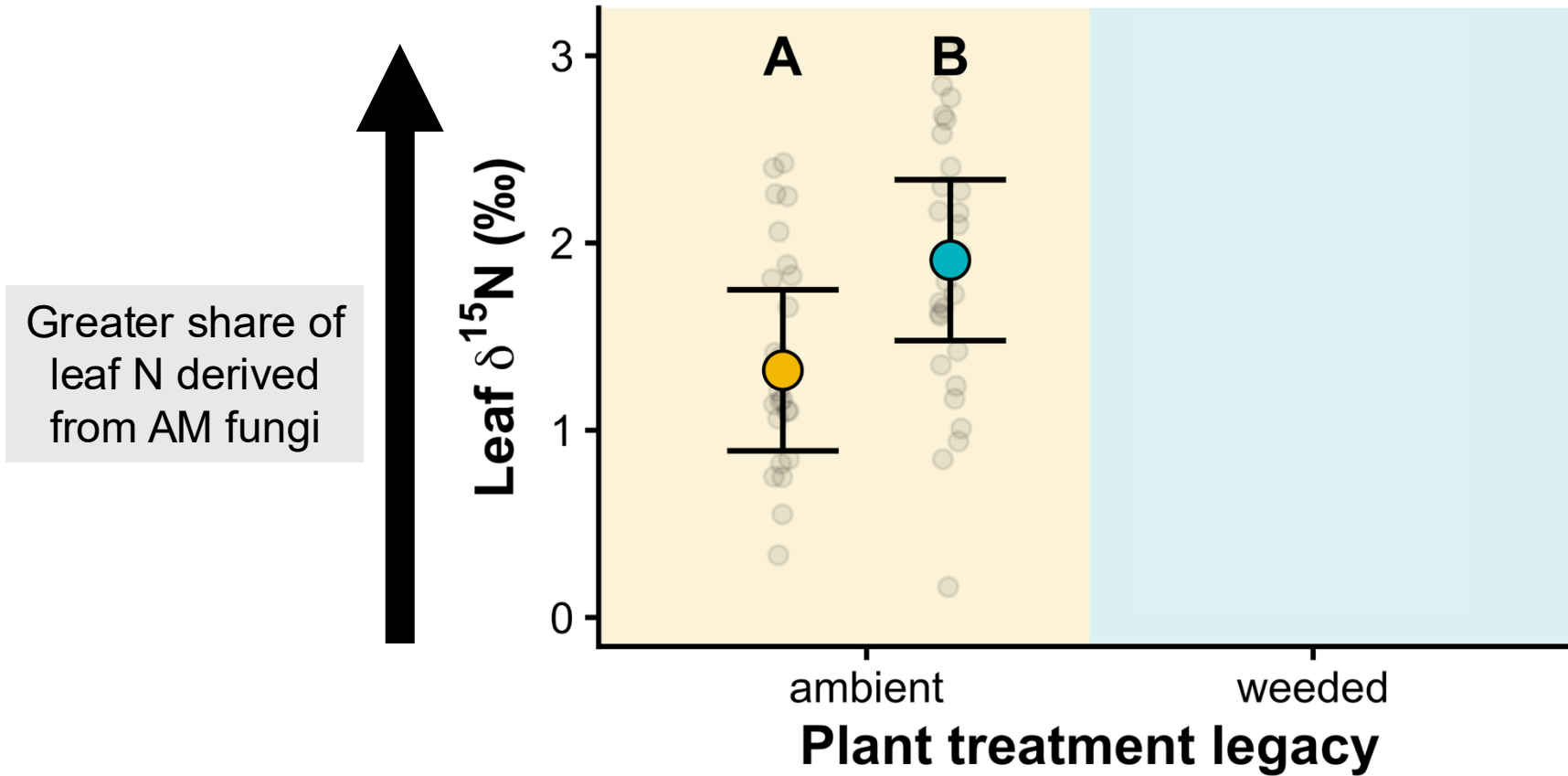


*only in plants with history of growing with garlic mustard

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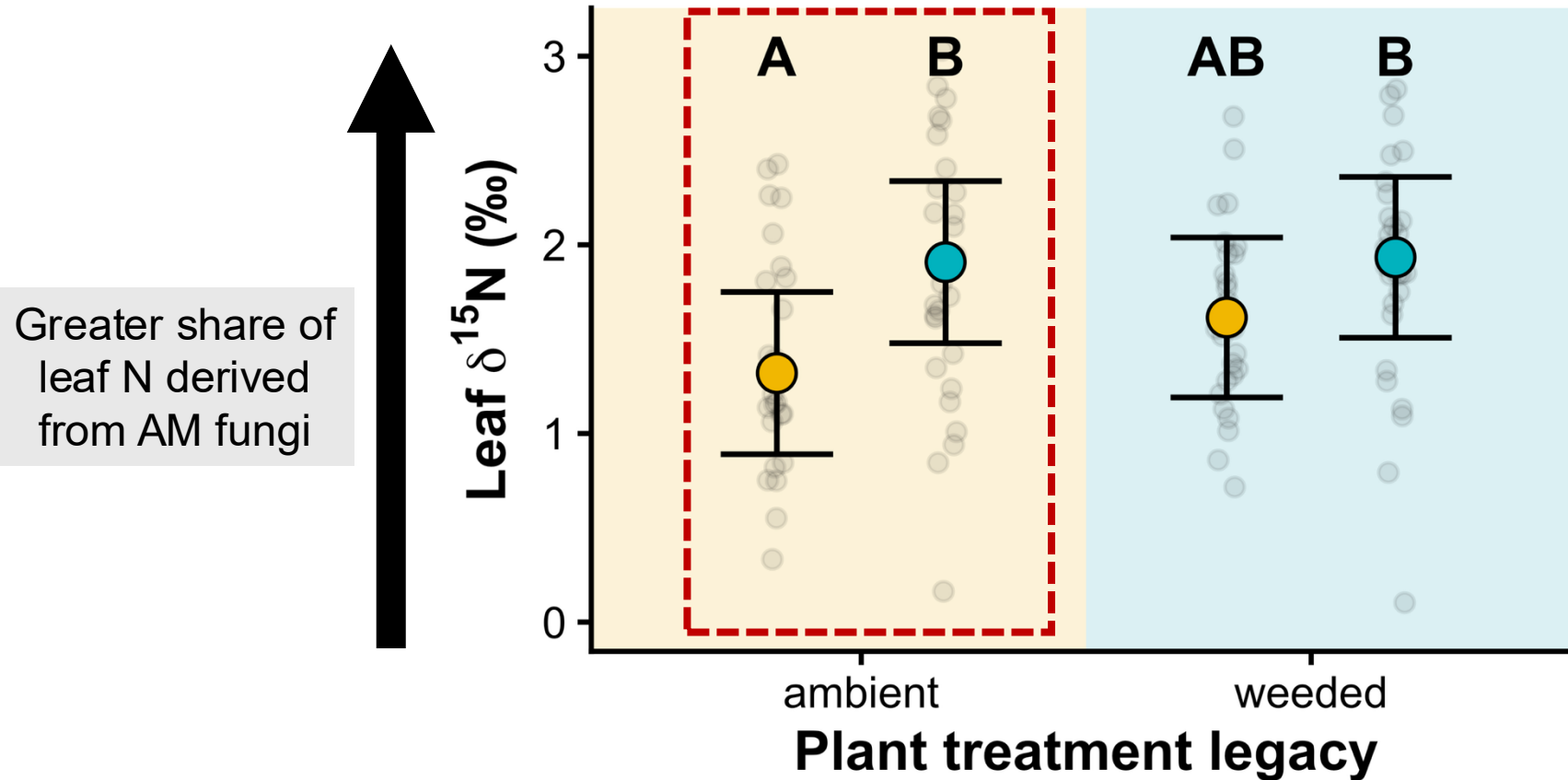
Ambient AM fungal source decreases AM-mediated plant N uptake



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Ambient AM fungal source decreases AM-mediated plant N uptake*



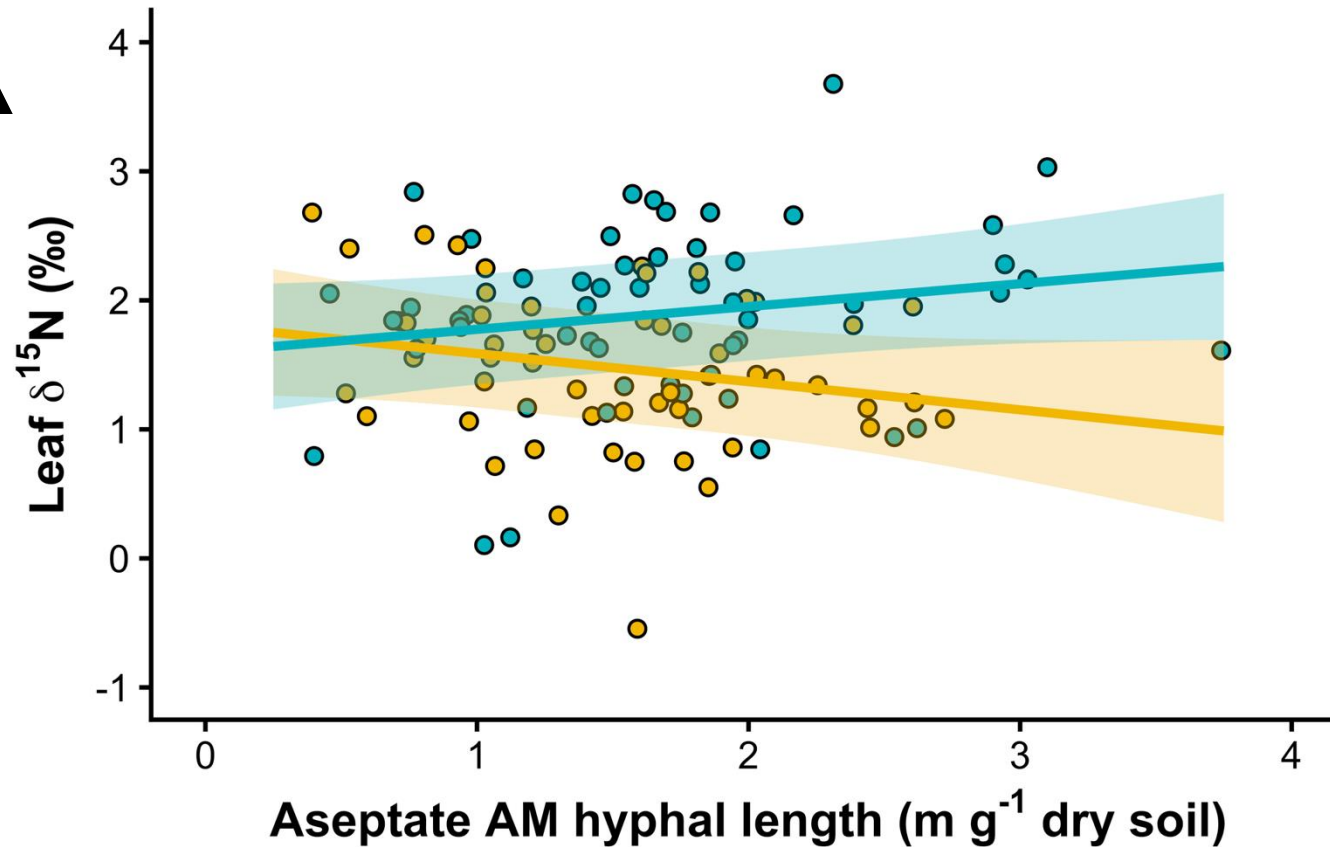
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AM hyphal length explains fungal source effects on AM-mediated plant N uptake

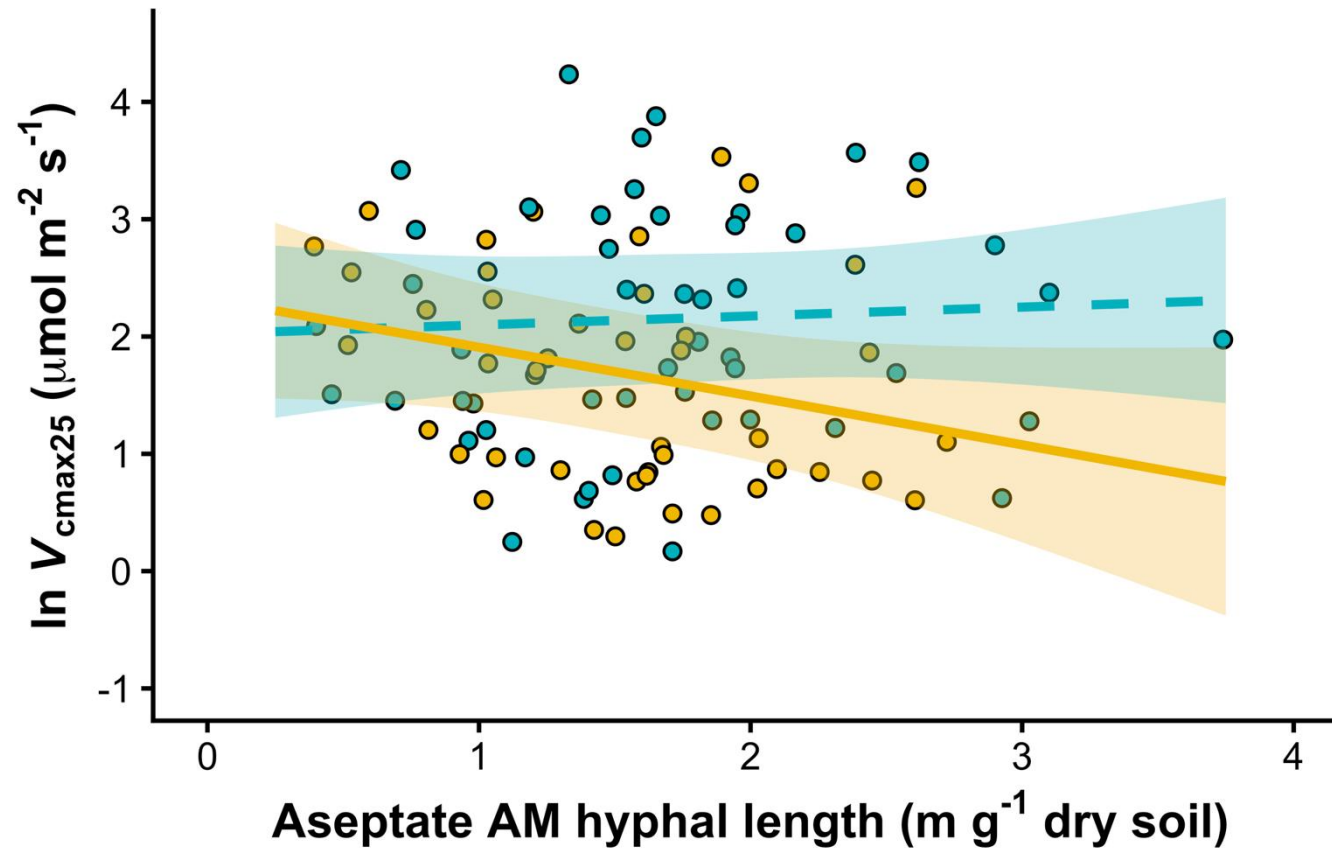
Greater share of
leaf N derived
from AM fungi



Ambient = GM present
Weeded = GM absent

AM fungal source ambient weeded

AM hyphal length explains fungal source effects on photosynthetic capacity



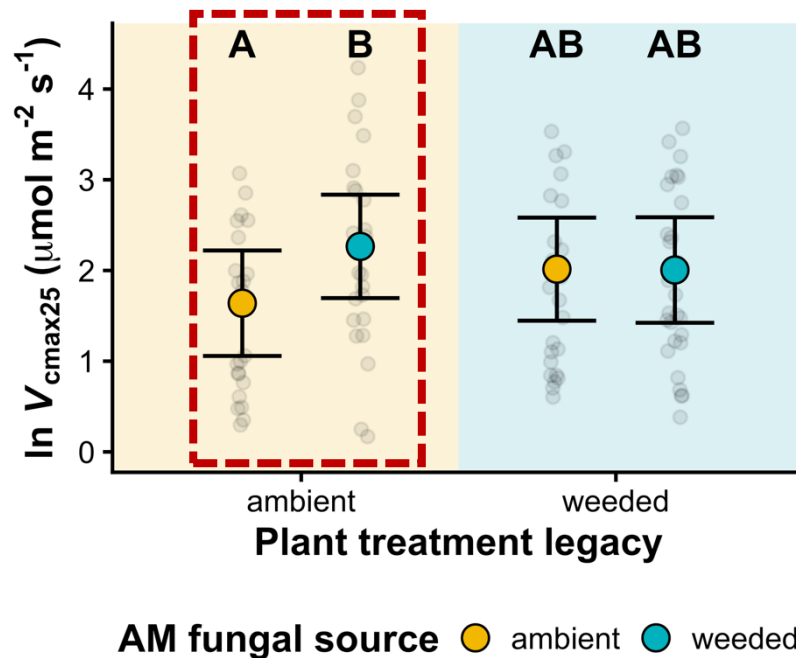
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Are photosynthetic responses to garlic mustard due to AM fungal mutualism disruption?

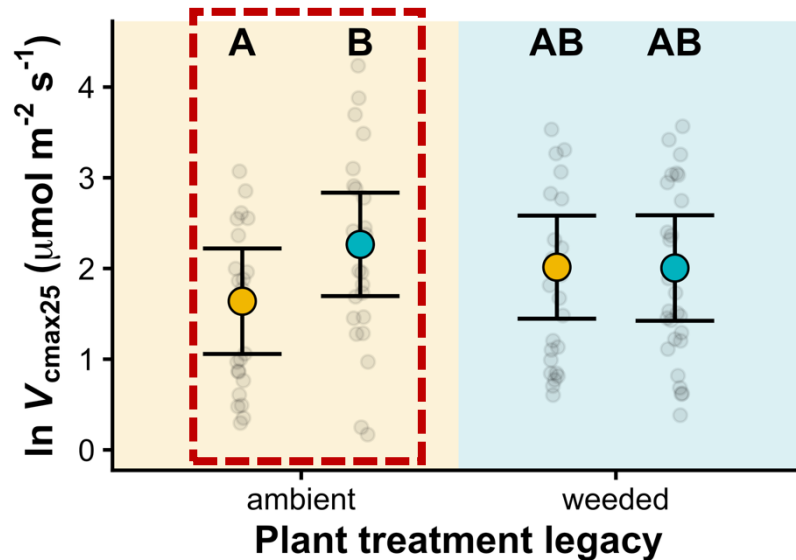
Are photosynthetic responses to garlic mustard due to AM fungal mutualism disruption?

YES

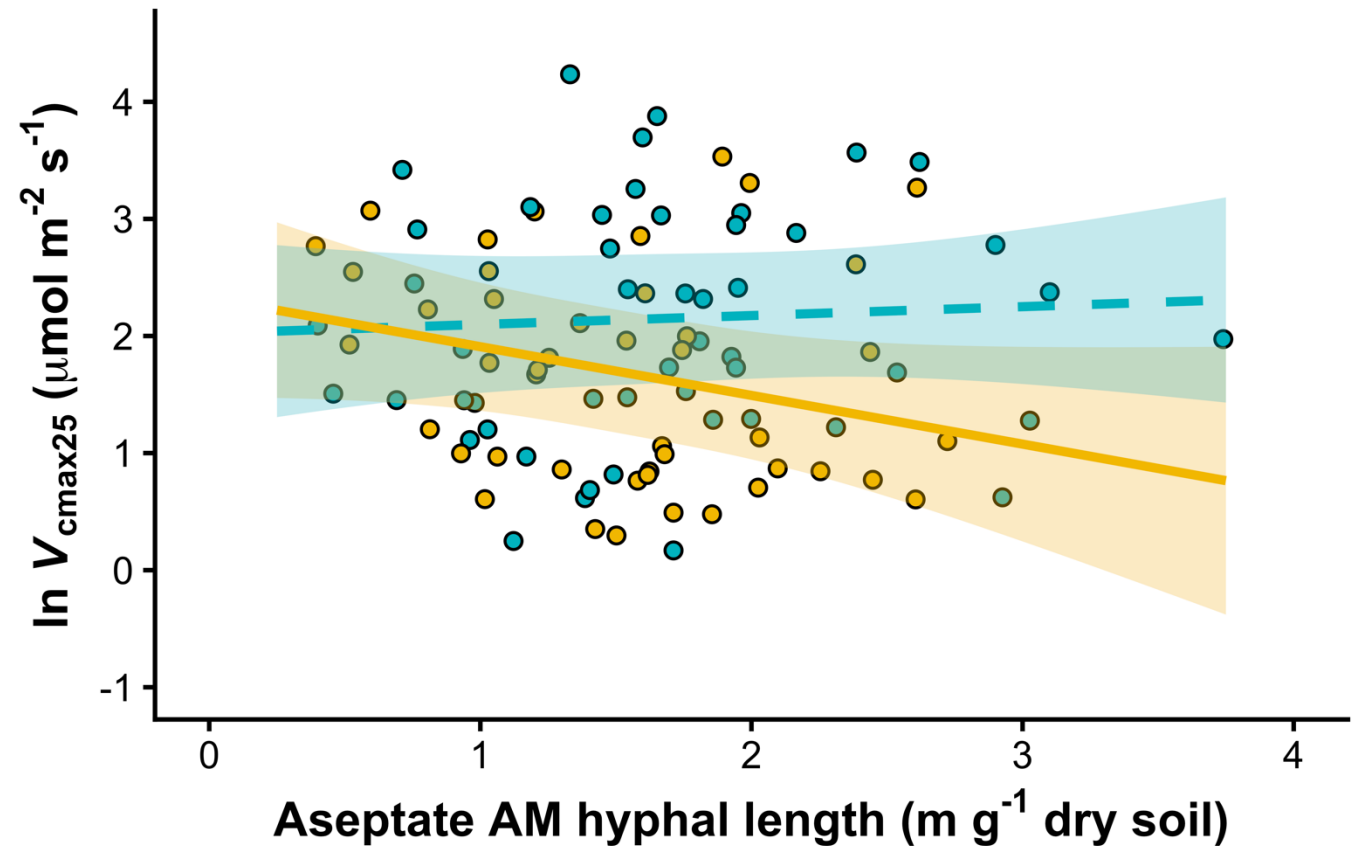


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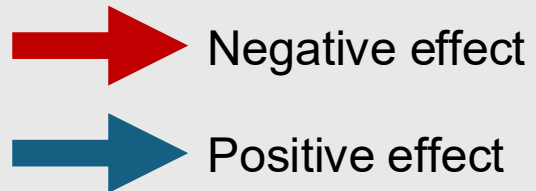
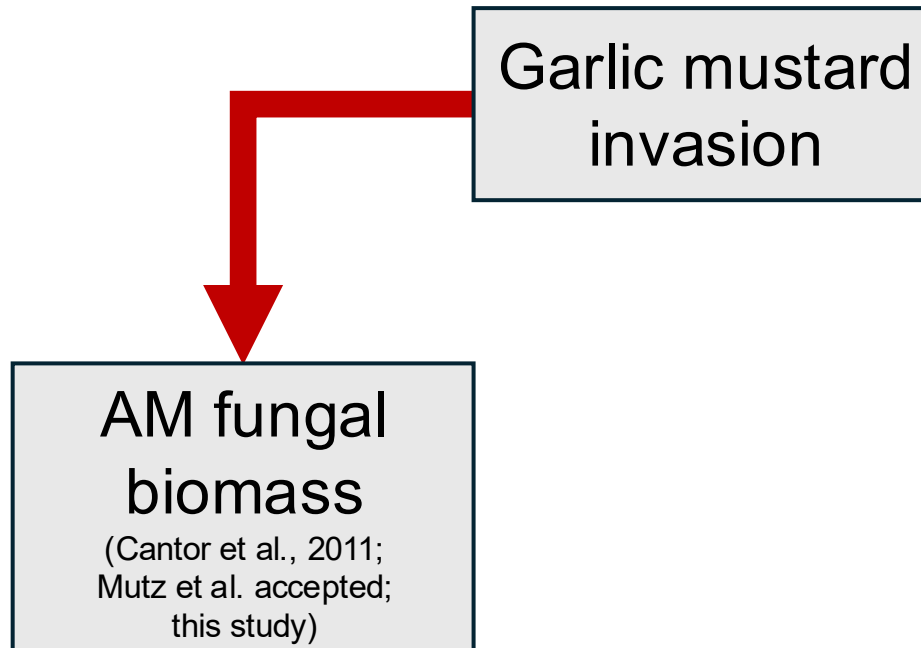
YES



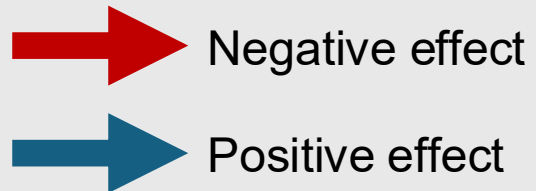
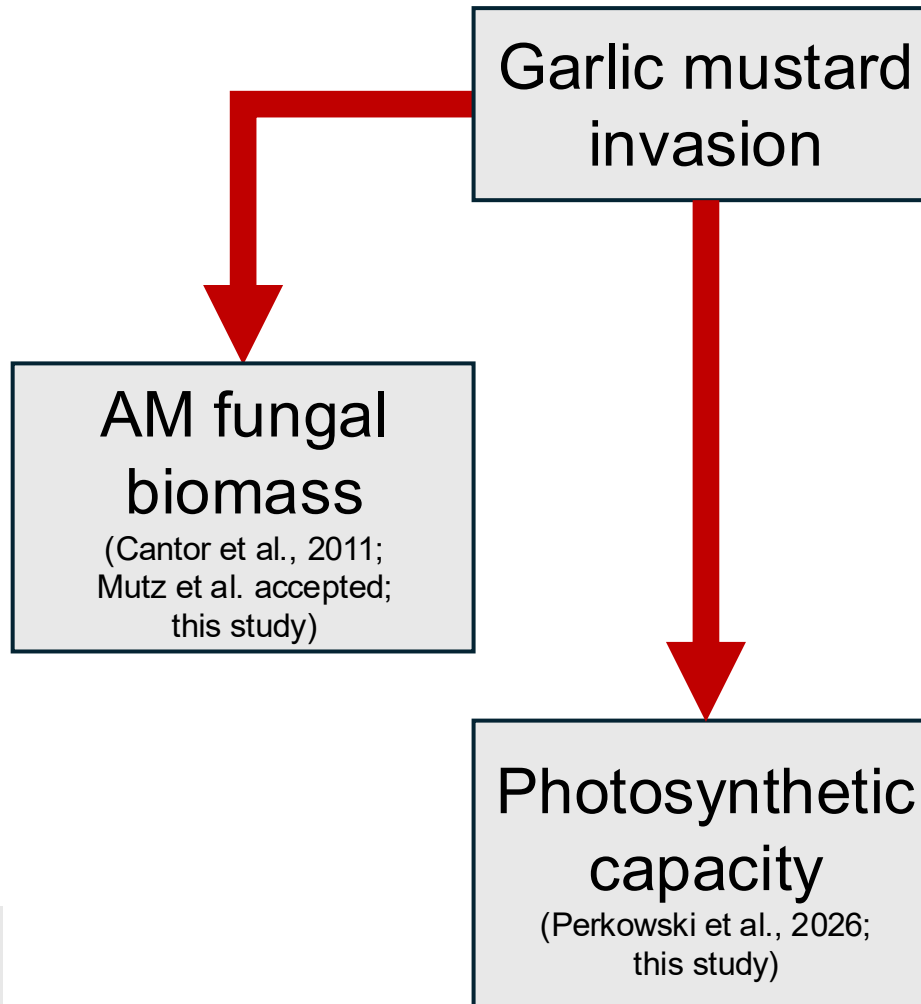
AM fungal source ● ambient ● weeded



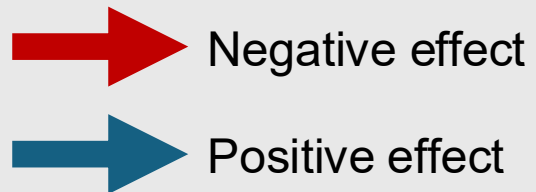
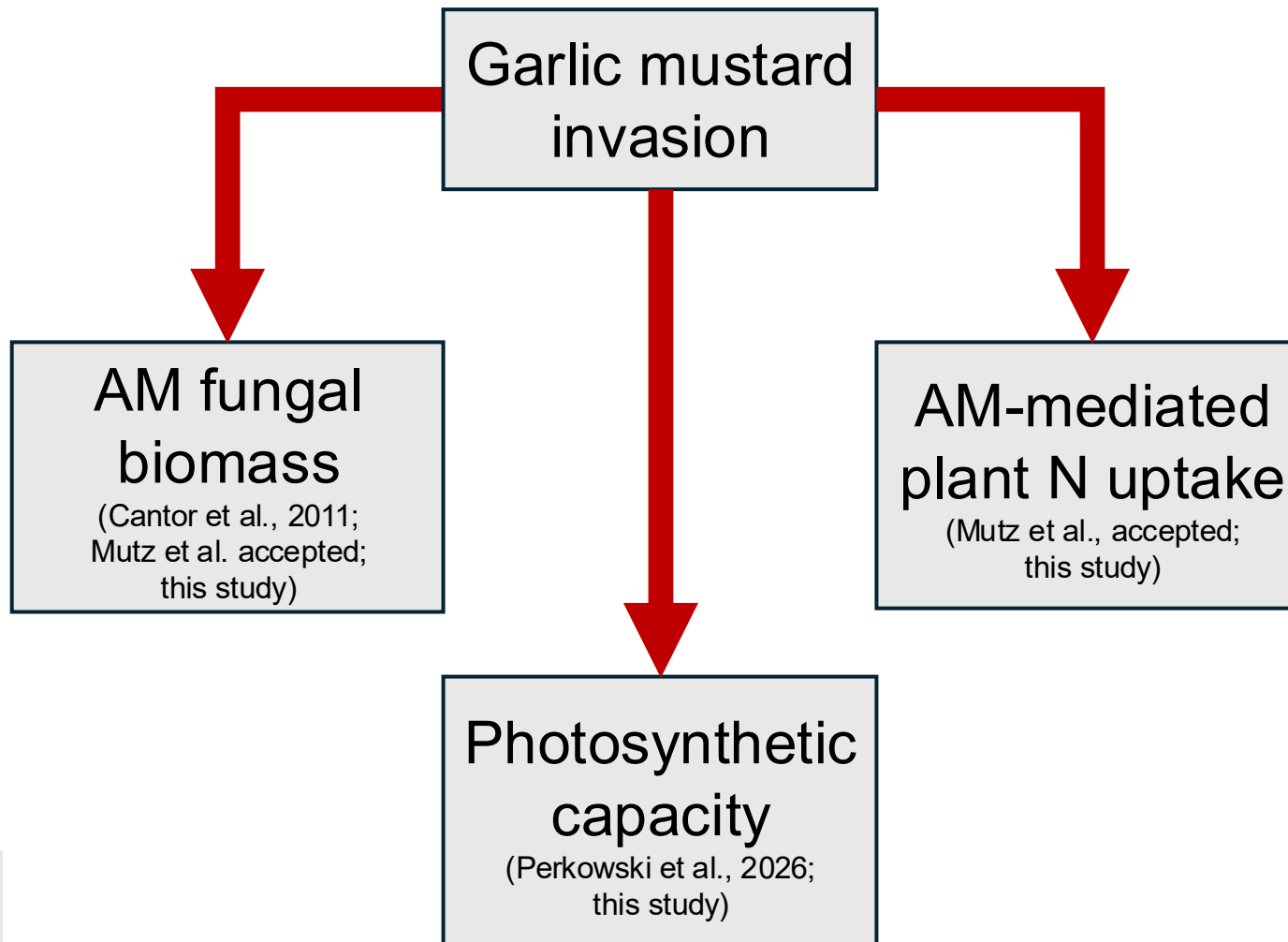
What is the mechanism that explains AM fungal effects on physiological responses to garlic mustard?



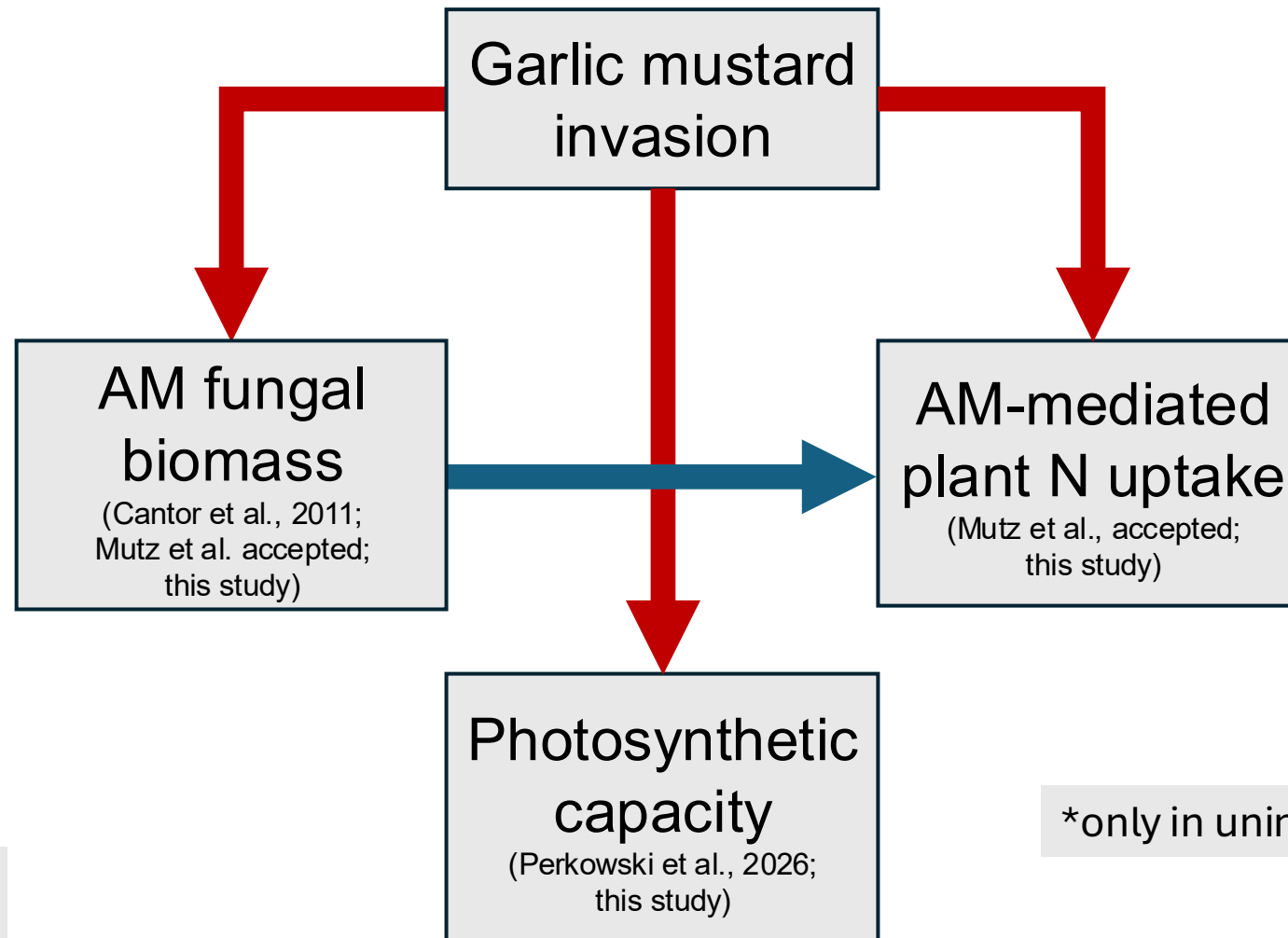
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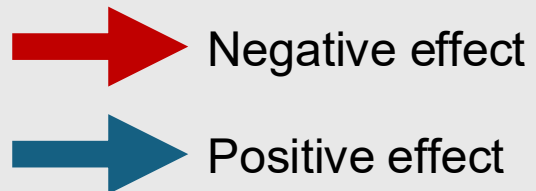
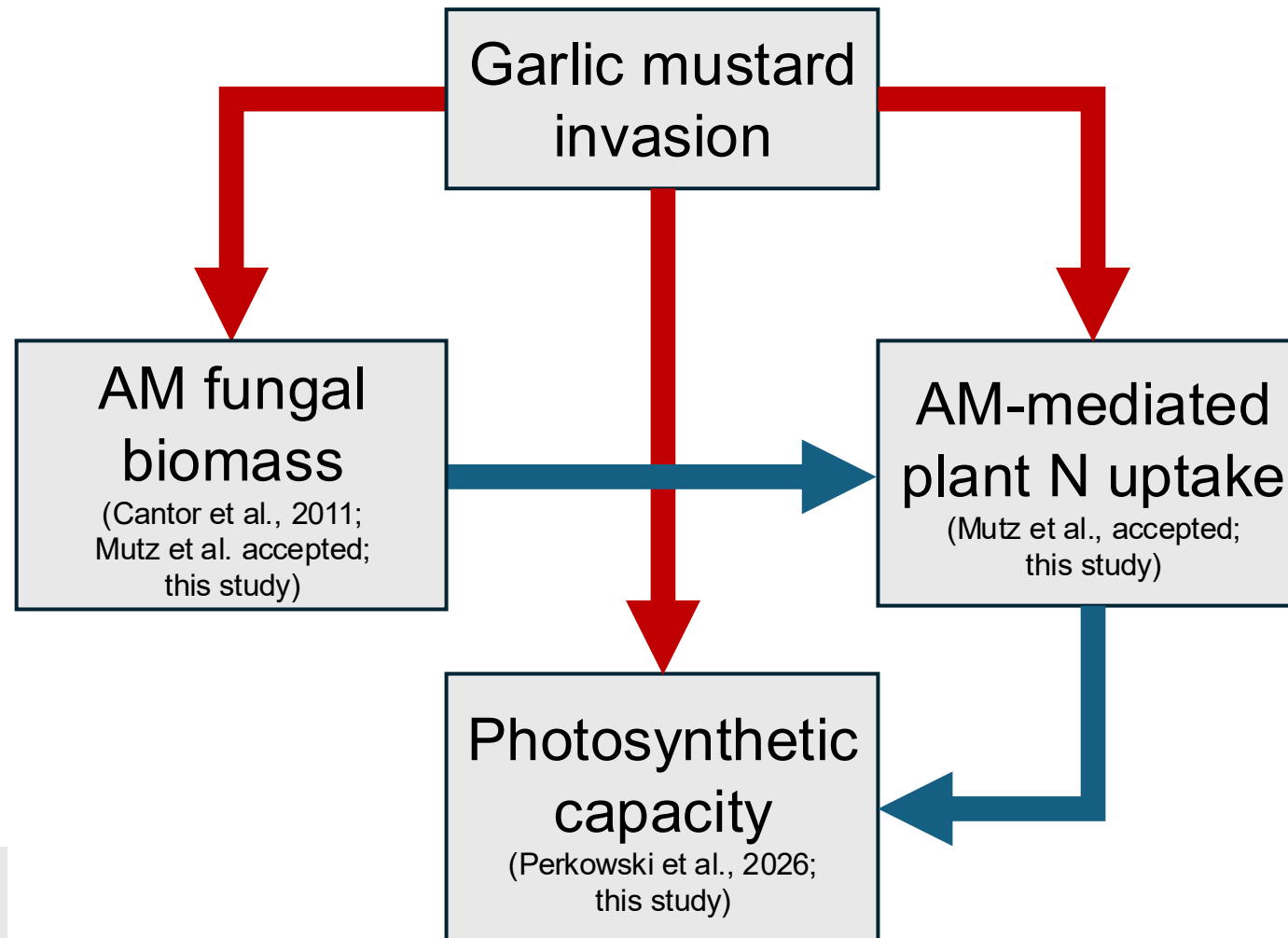
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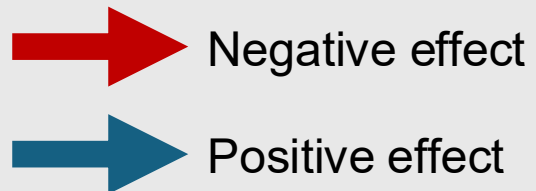
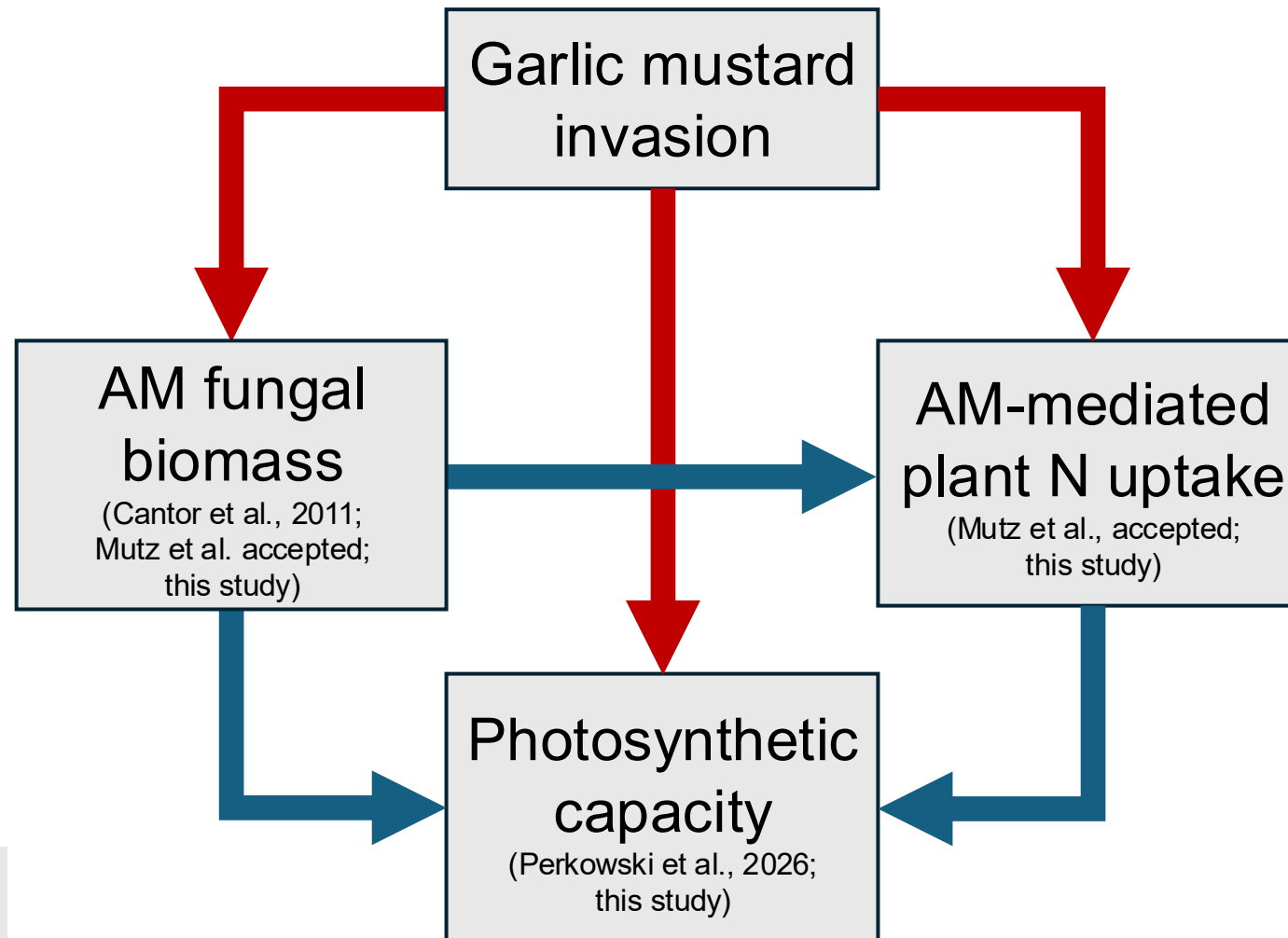
*only in uninvaded AM fungal source



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Thank you!

Please be sure to check out two undergraduate poster presentations related to this work at this afternoon's poster session:



Ava Payne
PS31-165: Biogeochemistry

“Refining split-pot designs to inform plant-mycorrhizal nutrient exchange dynamics”



Niyomi Meynari
PS31-167: Biogeochemistry

“The negative effects of garlic mustard on native plant physiology are rapidly restored after conditioning with uninvaded soils”